Infinite log-concavity of trinomial powers

A proposed computer-assisted proof of the all-exponent trinomial assertion

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Abstract

Let $a_{n,k} = [x^k](1+x+x^2)^n$, extended by zero outside its support, and define $(\mathcal{L}a)_k = a_k^2 - a_{k-1}a_{k+1}$. We give a computer-assisted proof that $\mathcal{L}^r a_n$ is entrywise nonnegative for every integer $n \geq 10$ and every integer $r \geq 0$. The proof joins an exact invariant-cone verification for $10 \leq n \leq 4003$ with a uniform analytic and algebraic argument for $n \geq 4004$. In the latter range, a sixteen-variable integral and explicit saddle estimates supply bulk signs for the fourth iterate; exact moving-node certificates supply the remaining edge signs. Palindromicity and a degree count then prove negative real-rootedness of the fourth iterate. Separate Toeplitz-minor arguments establish the earlier layers, and a real-rootedness preservation theorem gives every later layer.

The numerical components use exact integers and rational enclosures. The claimed quantifiers are not inferred from a finite sample of exponents. This draft has undergone independent internal checks, including separately implemented arithmetic checks; it has not undergone external peer review or proof-assistant formalization. No assertion of exhaustive literature priority is made.

The target and the proof architecture

Theorem. For every integer $n \geq 10$, every integer $r \geq 0$, and every integer k,

$$(\mathcal{L}^r a_n)_k \geq 0, \qquad a_{n,k} = [x^k](1 + x + x^2)^n.$$

All sequences and iterates use zero extension beyond their natural finite support.

This is the specific trinomial assertion discussed in Example 5.3 of Medina and Straub, *On multiple and infinite log-concavity*, arXiv:1405.1765v1, printed page 11; the published article appears in *Annals of Combinatorics* **20** (2016), 125–138. The original work verified a finite range $10 \leq n \leq 500$. The present theorem does not assert the more general polynomial conjecture considered elsewhere in that paper.

The proof has four logically separate components:

1. **Finite exponents.** Every $10 \leq n \leq 4003$ reaches a sufficient invariant cone under exact directed ratio arithmetic. The cone proves all subsequent iterations at once.
2. **Earlier layers in the infinite range.** Finite certificates at exponents eight and nine imply the full PF_4 property through symmetric-function identities. Convolution closure and determinant identities then prove nonnegativity through the third iterate for every $n \geq 56$.
3. **Fourth-iterate roots.** Uniform bulk estimates and exact certificates on the continuous parameter interval $0 \leq u \leq 1/4004$ provide enough alternating real signs to locate all $2n$ zeros of the fourth iterate on the negative real axis for every integer $n \geq 4004$.
4. **Later layers.** Brändén's preservation theorem applies to the fourth iterate and thereafter. It is not used backwards to infer positivity of the earlier layers.

The finite and infinite exponent ranges are complementary and leave no gap. The roles of the continuous parameter family and the finite polynomial at reciprocal integers are kept distinct throughout.

Verification conventions

A rational enclosure is a proved inclusion, not a floating-point approximation. Every strict sign assertion subtracts all applicable rounding and analytic remainder bounds before it is accepted.

The exact arithmetic backend is credited and pinned in the reproducibility package. A fast replay of saved enclosures and a full regeneration from the input formulas have different trust boundaries; the package documents both rather than conflating them.

The mathematical statements below are accompanied by portable verification commands and explicit certificate coverage. Raw development logs and internal review narratives are not part of the mathematical exposition.

Finite certificates and the first three iterates

Let $p(x) = 1 + x + x^2$ and $a_k = [x^k]p(x)^n$, with zero extension. All arithmetic certificates below use integers and directed rational enclosures.

1. Exact ratio verification for $10 \le n \le 4003$

The coefficient identity $p(x)(p(x)^n)' = np'(x)p(x)^n$ gives

$$(k+1)a_{k+1} = (n-k)a_k + (2n-k+1)a_{k-1}.$$

The coefficients are positive on $0 \le k \le 2n$ and symmetric. On the first half define $r_k = a_k/a_{k-1}$. Then $r_1 = n$ and

$$r_{k+1} = \frac{n-k+(2n-k+1)/r_k}{k+1}, \qquad 1 \le k < n.$$

This expression decreases with r_k. If $S = 2^{256}$ and $R_k^- \le Sr_k \le R_k^+$, directed integer divisions give

$$R_{k+1}^- = \left\lfloor \frac{(n-k)SR_k^+ + (2n-k+1)S^2}{(k+1)R_k^+} \right\rfloor,$$
$$R_{k+1}^+ = \left\lceil \frac{(n-k)SR_k^- + (2n-k+1)S^2}{(k+1)R_k^-} \right\rceil.$$

All denominators are positive, as checked by the certificate program.

For $q_k = a_{k-1}a_{k+1}/a_k^2$, the initial half-row enclosures are

$$Q_k^- = \left\lfloor \frac{SR_{k+1}^-}{R_k^+} \right\rfloor, \quad Q_k^+ = \left\lceil \frac{SR_{k+1}^+}{R_k^-} \right\rceil, \quad 1 \le k < n,$$
$$Q_n^- = \left\lfloor \frac{S^3}{(R_n^+)^2} \right\rfloor, \quad Q_n^+ = \left\lceil \frac{S^3}{(R_n^-)^2} \right\rceil,$$

and $Q_0^- = Q_0^+ = 0$. Symmetry gives the remaining indices. The powers of S in the central formula are essential.

Whenever $0 \le q_k < 1$, the next iterate is positive on its support and its ratios obey

$$q_k' = \frac{q_k^2(1-q_{k-1})(1-q_{k+1})}{(1-q_k)^2}.$$

The map increases in its own argument and decreases in either neighboring argument. Given integer enclosures $0 \le Q^- \le Sq \le Q^+ < S$, lower and upper bounds for Sq_k' are therefore

$$\left\lfloor \frac{(Q_k^-)^2(S-Q_{k-1}^+)(S-Q_{k+1}^+)}{S(S-Q_k^-)^2} \right\rfloor, \qquad \left\lceil \frac{(Q_k^+)^2(S-Q_{k-1}^-)(S-Q_{k+1}^-)}{S(S-Q_k^+)^2} \right\rceil.$$

At the center of the symmetric half-array the right neighbor equals the left neighbor. No unbounded integer is replaced by floating-point arithmetic.

The exact program checks every integer $n = 10, \dots, 4003$. At each step it verifies the strict upper bound $Q_k^+ < S$, and it terminates only when $3Q_k^+ \le S$ for every k. All 3994 cases terminate by depth 12. A separate verifier using exact coefficient rows instead of the successive-ratio initializer gives the same successful coverage. The terminal condition is sufficient because $q \le 1/3$ implies

$$q' \le \frac{(1/3)^2}{(1-1/3)^2} = \frac{1}{4} < \frac{1}{3}.$$

Induction proves positivity of all later iterates, so a finite exact certificate per exponent establishes infinite log-concavity rather than merely a bounded number of successful iterations.

2. Finite certificates for the full PF_4 property

For a zero-extended finite sequence a, let $T(a)_{i,j} = a_{j-i}$, $i, j \ge 0$. The property PF_4 means every minor of order at most four is nonnegative.

Suppose

$$A(x) = \sum_{k=0}^{d} a_k x^k = \prod_{h=1}^{d} (1 + z_h x),$$

so $a_k = e_k(z)$. It suffices to test the finite determinants

$$\det(a_{u_i - i + j})_{1 \le i,j \le 4}$$

for every $d \ge u_1 \ge u_2 \ge u_3 \ge u_4 \ge 0$. Here the z_h may be complex; the tests concern real evaluated determinants.

Indeed dual Jacobi–Trudi identifies these determinants with ordinary Schur functions $s_\lambda(z)$ for $\lambda' = u$, namely all shapes of width at most four and height at most d; zero padding includes smaller widths. Consider an arbitrary order-$\ell \le 4$ Toeplitz minor with increasing rows r_i and columns c_i. If $c_i < r_i$ for some i, the first i columns can meet at most $i - 1$ eligible rows, so every determinant term vanishes. Otherwise put

$$\lambda'_i = c_{\ell+1-i} + i, \qquad \mu'_i = r_{\ell+1-i} + i.$$

These are partitions with $\lambda' \ge \mu'$. Transposition and simultaneous reversal transform the minor into

$$\det(e_{\lambda'_i - \mu'_j - i + j}) = s_{\lambda/\mu}(z).$$

The Littlewood–Richardson expansion

$$s_{\lambda/\mu} = \sum_{\nu} c^{\lambda}_{\mu\nu} s_\nu$$

has nonnegative integer coefficients. For nonzero terms $\nu \subseteq \lambda$, hence $\mathrm{width}(\nu) \le \ell \le 4$. If $\mathrm{height}(\nu) > d$, its specialization to d variables vanishes. Every remaining ordinary Schur value is among the finite tests. Thus every arbitrary Toeplitz minor is nonnegative.

For $p(x)^n$ use $d = 2n$ and n copies each of $\exp(i\pi/3)$ and $\exp(-i\pi/3)$. The exact tests at $n = 8$ and $n = 9$ enumerate respectively $\binom{20}{4} = 4845$ and $\binom{22}{4} = 7315$ determinants. Every determinant is positive; the minimum is 1. These are finite certificates for the complete infinite-matrix PF_4 condition, not merely a sample of consecutive-row minors.

3. Convolution closure and nonnegative early iterates

We have $T(a * b) = T(a)T(b)$. Each fixed minor of the product has a finite Cauchy–Binet sum: contributing intermediate indices lie between the selected row and column indices. If both factors are PF_4, every summand is nonnegative. Thus convolution preserves PF_4.

For $n \geq 56$ write $n = 8q + r$, $0 \leq r \leq 7$. Since $q \geq 7 \geq r$,

$$n = 8(q - r) + 9r.$$

Hence p^n is a convolution of the two certified PF_4 seed rows, and is PF_4.

Let $D_j(k) = \det(a_{k+s-t})_{0 \leq s,t < j}$, $D_0 = 1$. Desnanot–Jacobi gives

$$D_j(k)^2 - D_j(k-1)D_j(k+1) = D_{j-1}(k)D_{j+1}(k).$$

It follows algebraically that

$$\begin{aligned}
(\mathcal{L}a)_k &= D_2(k), \\
(\mathcal{L}^2 a)_k &= a_k D_3(k), \\
(\mathcal{L}^3 a)_k &= D_2(k)\big[a_k^2 D_4(k) + D_3(k-1)D_3(k+1)\big].
\end{aligned}$$

These are polynomial identities with no division by a minor, so endpoints and possible zero minors cause no exception. The property PF_4 therefore implies nonnegativity of layers zero through three. The selected infinite range $n \geq 4004$ lies entirely within this proven range $n \geq 56$.

References

The symmetric-function identities are dual Jacobi–Trudi and Littlewood–Richardson; see I. G. Macdonald, *Symmetric Functions and Hall Polynomials*, second edition, Chapter I, Sections 3, 5 and 9. The first three iteration identities and fourth-order limitations are discussed by P. R. W. McNamara and B. E. Sagan, *Infinite log-concavity: developments and conjectures*, https://arxiv.org/abs/0808.1065. These general algebraic tools are prior results, not new claims of this proof.

Analytic part of a computer-assisted proof for powers of $1 + x + x^2$

1. Statement and structure of the proof

For a finitely supported sequence $a = (a_k)_{k\in\mathbb{Z}}$, extended by zero outside its support, write

$$(\mathcal{L}a)_k = a_k^2 - a_{k-1}a_{k+1}.$$

The sequence is **infinitely log-concave** if every iterate $\mathcal{L}^j a$ is nonnegative.

The result addressed by this draft is the following integer-exponent statement:

> **Theorem.** For every integer $n \geq 10$, the coefficient sequence of $(1+x+x^2)^n$ is infinitely log-concave.

This is the trinomial case discussed in Medina–Straub, Example 5.3 and Table 1. It is not the general assertion for every polynomial with positive coefficients in their Conjecture 5.2. See Medina–Straub, *On multiple and infinite log-concavity*.

The proof is computer-assisted in two explicit ways: certain low-degree rational polynomial inequalities are verified by positive Bernstein coefficients, and a finite collection of parameter-wide sign inequalities is verified by exact interval Taylor arithmetic. Floating-point calculations may select rational evaluation points, but do not certify their signs.

Put

$$p(z) = 1 + z + z^2, \qquad a_{n,k} = [z^k]p(z)^n, \qquad f_{n,j}(x) = \sum_k (\mathcal{L}^j a_n)_k x^k.$$

The analytic argument proves that $f_{n,4}$ has only negative real zeros for every integer $n \geq 4004$. Its degree is $2n$. We do not require, or assert here, this stronger real-rootedness statement throughout the smaller finite range.

The proof has three parts.

1. A uniform two-saddle estimate gives alternating signs of $f_{n,4}$ at explicitly defined bulk phase points.
2. Exact parameter-wide certificates supply a finite chain of edge signs and its connection to the bulk.
3. The intermediate value theorem and reciprocity exhaust the degree. The initial three iterates and the finite range $10 \leq n \leq 4003$ are treated in the companion finite and algebraic chapter above.

Once $f_{n,4}$ has negative real zeros, Brändén's preservation theorem applies repeatedly to the same zero-extended operator $\mathcal{L}$. The theorem and its proof are in §3 of Brändén, *Iterated sequences and the geometry of zeros*. It is used forward from the fourth iterate, not backward to infer the first three iterates.

Throughout, $\operatorname{sinc} x = \sin(x)/x$, with its continuous value at zero. Arguments of analytic functions are chosen continuously from the positive real axis whenever this is specified.

2. An exact sixteen-leaf integral

2.1. The diagonal identity

Consider a complete balanced binary tree with $r = 2^j$ labelled leaves. For an internal node ν, let L_ν, R_ν be its two child leaf sets, and put

$$Z_{\nu,L} = \prod_{i \in L_\nu} z_i, \qquad Z_{\nu,R} = \prod_{i \in R_\nu} z_i.$$

Define

$$P_j(z) = \prod_{\nu} \left(1 - \frac{Z_{\nu,L}}{Z_{\nu,R}}\right).$$

Then

$$f_{n,j}(x) = \mathrm{CT}_{z_1,\dots,z_{r-1}} P_j(z) \prod_{i=1}^{r} p(z_i)^n, \qquad z_r = \frac{x}{z_1 \cdots z_{r-1}}. \tag{2.1}$$

To prove this, start with the coefficientwise product formula for a diagonal. At one application of $\mathcal{L}$, the product of two equal-index coefficients is replaced by that product minus the product with indices shifted by $-1, +1$. These shifts insert exactly $Z_{\nu,L}/Z_{\nu,R}$. Induction down the binary tree proves (2.1), including zero extension at the boundaries.

For $x = -s^{16}$, take all sixteen radii equal to s, and write $z_i = se^{i\alpha_i}$. The angular constraint is

$$\sum_{i=1}^{16} \alpha_i = \pi \pmod{2\pi}.$$

The measure is normalized Haar measure on this fifteen-dimensional torus. In the coordinates $\alpha_1, \dots, \alpha_{15}$, it is $(2\pi)^{-15} d\alpha_1 \cdots d\alpha_{15}$.

2.2. A nonnegative kernel

For every internal node set

$$D_\nu(\alpha) = \sum_{i \in L_\nu} \alpha_i - \sum_{i \in R_\nu} \alpha_i.$$

The exponential factor $\prod_i p(se^{i\alpha_i})^n$ and the constrained measure are invariant under permutations of the leaves. Average the kernel over the group generated by interchanging the two children at each tree node. Beginning at the lowest level, an interchange changes D_ν to $-D_\nu$, while the already averaged descendant factors are even and merely permuted. It therefore replaces $1 - e^{iD_\nu}$ by $1 - \cos D_\nu$. Repeating upward gives the exact integral identity

$$f_{n,4}(-s^{16}) = \int_{\sum \alpha_i = \pi \ (2\pi)} K(\alpha) \prod_{i=1}^{16} p(se^{i\alpha_i})^n \, d\mu(\alpha), \qquad K = \prod_{\nu} (1 - \cos D_\nu) \geq 0. \tag{2.2}$$

The replacement is an equality after integration; it is not a pointwise equality with the original Laurent kernel.

At a configuration of equal angles, the leading homogeneous term of K is

$$H(\eta) = \prod_{\nu} \frac{D_\nu(\eta)^2}{2}. \tag{2.3}$$

There are fifteen internal nodes, so this vanishing has degree thirty.

2.3. Haar coordinates and measure normalization

On the tangent space

$$\mathcal{T} = \{\eta \in \mathbb{R}^{16} : \sum_i \eta_i = 0\},$$

the fifteen contrast vectors defining D_ν are orthogonal. Their squared norms are the node sizes m_ν. Sizes $2, 4, 8, 16$ occur respectively $8, 4, 2, 1$ times. For orthonormal coordinates $x_\nu = D_\nu/\sqrt{m_\nu}$,

$$H = \left(\prod_\nu \frac{m_\nu}{2}\right) \prod_\nu x_\nu^2 = 2048 \prod_\nu x_\nu^2. \tag{2.4}$$

The original fifteen angular coordinates have metric matrix $I + \mathbf{1}\mathbf{1}^T$, of determinant sixteen. Thus their measure is one quarter of Euclidean measure on $\mathcal{T}$. This factor is important in every saddle and complement estimate below.

3. Saddle geometry and explicit scalar inequalities

Set

$$\theta = \frac{\pi}{16}, \quad \zeta = se^{i\theta}, \quad q(z) = 1 + 4z + z^2, \quad m(z) = \frac{zp'(z)}{p(z)}, \quad v(z) = \frac{zq(z)}{p(z)^2}, \quad V = nv(s).$$

For $m \geq 2$, define

$$\kappa_m(z) = \left(z\frac{d}{dz}\right)^{m-2} v(z).$$

Here $v(s) > 0$ for $s > 0$. We work with $0 < s \leq 1$; reciprocity later supplies the other half of the zeros.

3.1. Modulus estimates

Write

$$h_s(\alpha) = \log \frac{|p(se^{i\alpha})|}{p(s)}.$$

If $c = \cos\alpha$, the exact identity is

$$\frac{|p(se^{i\alpha})|^2}{p(s)^2} = 1 - D(c), \quad D(c) = 2a(1-c) + b(1-c^2),$$

where

$$a = \frac{s(1+s^2)}{p(s)^2}, \qquad b = \frac{4s^2}{p(s)^2}, \qquad a + b = v(s), \qquad b \leq \frac{2}{3}v(s). \tag{3.1}$$

In particular, if $|\alpha| \geq \pi/3$,

$$h_s(\alpha) \leq -\frac{5}{12}v(s). \tag{3.2}$$

A small-angle bound at the equal-phase saddle is

$$16h_s(\theta) > -\frac{40}{123}v(s). \tag{3.3}$$

For example, $D(\cos\theta) \leq v(s)\theta^2$, $v(s) \leq 1$, and $-\log(1-x) \leq x/(1-x)$ give

$$-16h_s(\theta) \leq \frac{\pi^2}{32(1-\pi^2/256)}v(s) < \frac{40}{123}v(s).$$

On the central interval,

$$\Re v(se^{i\alpha}) \geq \frac{1}{3}v(s), \qquad |\alpha| \leq \frac{\pi}{3}. \tag{3.4}$$

Consequently $h_s''(\alpha) = -\Re v(se^{i\alpha}) \leq -v(s)/3$: the function is strictly concave there.

These facts locate the two modulus maxima. A configuration containing a noncentral angle has total modulus loss at least $5v(s)/12$, larger than the equal-phase loss in (3.3). If all angles are central, the possible winding sums are $\pm\pi, \pm 3\pi, \pm 5\pi$. Jensen's inequality shows that the largest modulus in each such sector occurs at its equal-angle configuration. Since h_s is even and strictly decreasing on $(0, \pi/3]$, the global maxima are exactly the two configurations with all angles equal to θ or $-\theta$.

3.2. Scalar certificate lemma

The following bounds hold uniformly for $0 < s \leq 1$. Put $v_0 = v(s)$, $\beta = \Re v(\zeta)$.

$$\frac{19}{20}v_0 \leq \beta \leq \frac{17}{16}v_0, \qquad \left(\frac{|v(\zeta)|}{\beta}\right)^{45/2} < 2, \tag{3.5}$$

$$|\Im\kappa_3(\zeta)| \leq \frac{1}{4}v_0, \quad |\kappa_3(\zeta)| \leq \frac{6}{5}v_0, \quad |\kappa_4(\zeta)| \leq \frac{4}{3}v_0, \quad |\kappa_5(\zeta)| \leq \frac{9}{4}v_0. \tag{3.6}$$

On the wider sector $\cos\alpha \geq 3/4$,

$$\Re\kappa_4(se^{i\alpha}) \leq \frac{3}{2}v_0, \quad |\kappa_4(se^{i\alpha})| \leq 3v_0, \quad |\kappa_6(se^{i\alpha})| \leq 32v_0. \tag{3.7}$$

These are rational polynomial inequalities, rather than numerical bounds on a grid. Here is an explicit way to verify them. If $\kappa_m(z) = N_m(z)/p(z)^m$, then

$$N_2 = zq(z), \qquad N_{m+1} = z\{pN_m' - mp'N_m\}. \tag{3.8}$$

For a polynomial $N(z) = \sum a_i z^i$,

$$|N(se^{i\alpha})|^2 = \sum_{i,j} a_i a_j s^{i+j} T_{|i-j|}(\cos\alpha), \tag{3.9}$$

where T_j is a Chebyshev polynomial. Clearing positive denominators in each squared modulus bound produces a polynomial in s, c. On the narrow sector, use $c \in [49/50, 1]$, containing $\cos\theta$, and split $s \in [0, 1]$ into four equal intervals. On the wide sector use $c \in [3/4, 1]$. Nonnegative tensor-product Bernstein controls verify the required inequalities. For a power-basis polynomial $\sum a_{ij}x^i y^j$ of bidegree (d, e), the controls are explicitly

$$b_{kl} = \sum_{i\leq k,\, j\leq l} a_{ij} \frac{\binom{k}{i}}{\binom{d}{i}} \frac{\binom{l}{j}}{\binom{e}{j}}. \tag{3.10}$$

Thus the verification uses rational arithmetic only.

For completeness, some especially useful reductions are small. Let $t = s + s^{-1}$, $c = \cos\alpha$,

$$A = tc + 1, \quad B^2 = (t^2 - 4)(1 - c^2), \quad D = A^2 + B^2, \quad R = (A + 3)A^2 + (A - 3)B^2.$$

Then $\Re v = R/D^2$ and $v(s) = (t+4)/(t+1)^2$. The polynomial

$$R(t+1)^2 - \tfrac{1}{3}(t+4)D^2$$

has positive Bernstein controls after $t = 2+u$, $c = (1+y)/2$: the minima by powers $u^0, \dots, u^5$ are

$$148, \quad 347/3, \quad 65/3, \quad 1/2, \quad 2/3, \quad 1/6.$$

This proves (3.4). The conditioning inequality follows by verifying $16(\Im v)^2 < (\Re v)^2$; hence $(|v|/\Re v)^{45/2} < (17/16)^{45/4} < 2$. The lower real-part bound in (3.5) and the bounds involving real or imaginary parts in (3.6)–(3.7) are verified after the same denominator clearing, using $t \geq 2$ where convenient. The upper real-part bound also follows directly by integrating $|\kappa_4| \leq 3v_0$ twice from angle zero:

$$\beta \leq v_0(1 + 3\theta^2/2) < 17v_0/16.$$

The accompanying exact certificate routines specify all polynomial subdivisions; their role is confined to these finite rational inequalities.

4. Leading amplitude and first correction

Near the positive saddle write $\alpha_i = \theta + \eta_i$, $\eta \in \mathcal{T}$. The exponent has quadratic part

$$-\frac{nv(\zeta)}{2}\sum_i \eta_i^2.$$

Using (2.4), the complex Gaussian integral is a product of fifteen one-dimensional second-moment integrals. Including the factor $1/4$ from the measure gives the one-saddle amplitude

$$A_n(s) = \frac{2048}{4(2\pi)^{15/2}} \frac{p(\zeta)^{16n}}{(nv(\zeta))^{45/2}}. \tag{4.1}$$

The square root and power are chosen continuously with $\Re v(\zeta) > 0$. The negative saddle contributes the complex conjugate.

Let $S_j = \sum_i \eta_i^j$ and $Q = \sum_\nu D_\nu^2$. The first correction to the Gaussian integrand is

$$-\frac{Q}{12} + \frac{n\kappa_4(\zeta)}{24}S_4 - \frac{n^2\kappa_3(\zeta)^2}{72}S_3^2. \tag{4.2}$$

The relevant normalized unit-scale weighted Gaussian moments are

$$\mathbb{E}Q = 192, \qquad \mathbb{E}S_4 = 5055/16, \qquad \mathbb{E}S_3^2 = 2835/4.$$

They yield

$$B_1(s) = -\frac{16}{v(\zeta)} + \frac{1685\kappa_4(\zeta)}{128v(\zeta)^2} - \frac{315\kappa_3(\zeta)^2}{32v(\zeta)^3}. \tag{4.3}$$

Equations (3.5)–(3.6) imply

$$\left|\frac{B_1(s)}{n}\right| \leq \frac{2173895}{41154V} < \frac{53}{V}. \tag{4.4}$$

No approximation to the zeros is inferred from this formal correction alone. The next two sections supply actual remainder bounds.

5. A uniform local remainder

Let

$$\mathcal{B} = \{\eta \in \mathcal{T} : |\eta_i| \le 2/5 \text{ for all } i\}.$$

We compare the exact integral over this box to the Gaussian plus (4.2) over the **same** box. The resulting one-saddle error is

$$\left|I_{\mathcal{B}} - I_{\mathcal{B}}^{(1)}\right| \le \frac{36821}{V^2}|A_n(s)|, \qquad V \ge 100. \tag{5.1}$$

Here the factors $p(\zeta)^{16n}$ and angular normalization are included in both integrals.

5.1. An even homotopy

Put

$$L(\eta) = \sum_i \{\log p(\zeta e^{i\eta_i}) - \log p(\zeta)\}, \quad E_\tau = nL(\tau\eta)/\tau^2,$$

$$F_\tau = \prod_\nu \operatorname{sinc}^2(\tau D_\nu/2), \qquad J(\tau,\eta) = HF_\tau e^{E_\tau}.$$

All singularities at $\tau = 0$ are removable. In particular, $E_0 = -nv(\zeta)\|\eta\|^2/2$. Since $J(-\tau,\eta) = J(\tau,-\eta)$, integration over the symmetric box is even in τ. Taylor's integral remainder therefore gives exactly

$$\int_{\mathcal{B}} \{J(1) - J(0) - \tfrac{1}{2}J''(0)\} = \int_0^1 \frac{(1-\tau)^3}{6} \int_{\mathcal{B}} \partial_\tau^4 J(\tau,\eta)\, d\eta\, d\tau. \tag{5.2}$$

Parity is used only in this integrated identity. Odd terms in the fourth derivative cannot be discarded at nonzero τ, because e^{E_τ} need not be even in η.

5.2. Bounds for the fourth derivative

For $m \ge 1$, Taylor's integral formula gives

$$\partial_\tau^m E_\tau = n \sum_i (i\eta_i)^{m+2} \int_0^1 u^m (1-u)\kappa_{m+2}(\zeta e^{iu\tau\eta_i})\, du. \tag{5.3}$$

Set $d = 2/5$, $A = |S_3|$, $B = S_4$, $C = |S_5|$, $D_6 = S_6$. The scalar bounds imply

$$\begin{aligned} e_1 &= V\{A/5 + (\tau/9 + 9d\tau^2/160 + 8d^2\tau^3/45)B\}, \\ e_2 &= V(1/9 + 9d\tau/80 + 8d^2\tau^2/15)B, \\ e_3 &= V(9C/80 + 16\tau D_6/15), \\ e_4 &= 16VD_6/15, \end{aligned} \tag{5.4}$$

with $|\partial_\tau^m E_\tau| \le e_m$. In the first two lines only, use $|S_5| \le dS_4$, $S_6 \le d^2 S_4$.

The identity $\operatorname{sinc}^2(t/2) = \mathbb{E}e^{it(U+U')}$, for independent uniforms on $[-1/2, 1/2]$, gives

$$|F| \le 1, \quad |F'| \le \tau Q/6 = f_1, \quad |F''| \le Q/6 = f_2,$$

$$|F'''| \le \tau Q^2/12 = f_3, \qquad |F''''| \le Q^2/12 = f_4. \tag{5.5}$$

For example, the exact fourth moment is $Q^2/12 - \sum D_\nu^4/60$, so dropping the last term gives an upper bound. The odd derivative estimates follow by integrating the next derivative from zero.

The product rule now gives

$$|J^{(4)}| \le He^{\Re E_\tau}\mathcal{P},$$
$$\begin{aligned}\mathcal{P} = {} & f_4 + 4f_3e_1 + 6f_2(e_2+e_1^2) + 4f_1(e_3+3e_1e_2+e_1^3)\\ & + e_4 + 4e_1e_3 + 3e_2^2 + 6e_1^2e_2 + e_1^4.\end{aligned} \tag{5.6}$$

This polynomial has nonnegative coefficients in its nonnegative arguments.

5.3. Gaussian domination and a reproducible finite moment sum

Real-part Taylor bounds give

$$\Re E_\tau \le -\frac{n}{2}a\|\eta\|^2, \qquad a = \beta - \frac{4}{75}v_0 \ge \frac{269}{300}v_0 > \left(\frac{15}{16}\right)^2 v_0.$$

The ratio of this real Gaussian normalization to the absolute complex Gaussian normalization is

$$(|v(\zeta)|/a)^{45/2} < 2(285/269)^{45/2} < 8. \tag{5.7}$$

Under the normalized H-weighted Gaussian at unit scale, the orthonormal Haar coordinates are independent with density $x^2e^{-x^2/2}/\sqrt{2\pi}$. Hence

$$\mathbb{E}D_\nu^{2r} = (2r+1)!!m_\nu^r, \qquad \mathbb{E}D_\nu^{2r+1} = 0. \tag{5.8}$$

The reconstruction

$$\eta_i = \sum_{\nu \text{ ancestor of } i} \frac{\epsilon_{i,\nu}D_\nu}{m_\nu}, \qquad \epsilon_{i,\nu} \in \{-1,1\}, \tag{5.9}$$

makes every moment needed in (5.6) a finite rational sum. Absolute odd powers of S_3 or S_5 are bounded by Cauchy–Schwarz, moving one odd factor to each side so that both resulting powers are even. An integer ceiling of the resulting square root is used. For example,

$$\mathbb{E}S_3^4 = 38064735/16, \quad \mathbb{E}S_5^2 = 81661725/256, \quad \mathbb{E}S_4^4 = 2166032330330625/65536.$$

For a monomial $C_0\tau^pV^lA^aB^bC^cD_6^dQ^e$ in (5.6), let $q = 3a+4b+5c+6d+2e$ denote its angular degree. Every monomial has $q - 2l \ge 4$. Its contribution to the coefficient of V^{-2}, for $V \ge 100$, is bounded by

$$8\,\frac{p!}{(p+4)!}\,[\text{monomial coefficient}]\,[\text{unit moment bound}]\left(\frac{16}{15}\right)^q 10^{2l-q+4}. \tag{5.10}$$

Here $p!/(p+4)!$ is the exact integral of $\tau^p(1-\tau)^3/6$; replacing all these weights by $1/24$ is unnecessarily wasteful.

Expanding the explicit polynomial (5.6), applying (5.8)–(5.10), and adding rational upper bounds gives

$$\frac{923292840775960455297735843730609930 24}{250758129586670547723770141601 5625} < 36821.$$

This proves (5.1). The moment calculation is finite and exact; it is not an asymptotic assertion with an unspecified constant.

5.4. Restoring the Gaussian model outside the box

The exact leaf-coordinate moment-generating function under the normalized real weighted Gaussian is

$$\mathbb{E}e^{t\sqrt{n\beta}\eta_i} = e^{15t^2/32} \prod_{m=2,4,8,16} (1 + t^2/m). \tag{5.11}$$

Indeed a leaf has four ancestor Haar coordinates with squared coefficients $1/2, 1/4, 1/8, 1/16$. A union and Chernoff bound therefore give

$$P_{\mathcal{B}^c} \le 32e^{-td\sqrt{19V/20}+15t^2/32} \prod_{m=2,4,8,16} (1 + t^2/m). \tag{5.12}$$

The absolute value of (4.2) is at most

$$Q/12 + (V/18)S_4 + (V^2/50)S_3^2.$$

Its weighted L^2 norm is bounded by $67/V$; the exact squared constant is

$$7405704307825/1693651716 < 67^2.$$

Cauchy–Schwarz and the conditioning factor in (3.5) bound the restored Gaussian-plus-correction tail by

$$2\{P_{\mathcal{B}^c} + (67/V)\sqrt{P_{\mathcal{B}^c}}\}|A_n(s)|.$$

For $V \ge 225$, choose $t = 5$ in (5.12). Exact rational exponential and square-root bounds give

$$P_{\mathcal{B}^c} < 1/1000, \qquad 2\{P_{\mathcal{B}^c} + (67/V)\sqrt{P_{\mathcal{B}^c}}\} < 11/500. \tag{5.13}$$

These are tails of the Gaussian model. They are not substituted for the actual torus complement, which is bounded next.

6. The actual torus complement at $V \ge 250$

Represent each angle in $[-\pi, \pi]$; boundary sets have measure zero. Call an angle central if $|\alpha_i| \le \pi/3$. We divide the complement of the two local boxes into five classes. All bounds in this section are relative to the magnitude of **one** saddle, $|A_n(s)|$.

6.1. Principal central sectors outside the boxes

In the sector $\sum \alpha_i = \pi$, set $\eta_i = \alpha_i - \theta$, and define

$$g(x) = \frac{h_s(\theta) + h_s'(\theta)x - h_s(\theta + x)}{v_0}.$$

For either radial direction, the scalar inequalities imply at $d = 2/5$

$$g(\pm d) \ge 269/3750, \qquad \frac{d}{dr} g(\pm r)\Big|_{r=d} \ge 43/125.$$

These follow by integrating $g''(0) \ge 19/20$, $|g'''(0)| \le 1/4$, and $g'''' \ge -3/2$. For larger radial distances still inside the central angular interval, (3.4) gives $g'' \ge 1/3$. Thus

$$g(\pm r) \ge 269/3750 + (43/125)(r - d) + (r - d)^2/6 \quad (r \ge d).$$

Let

$$\gamma(R) = \frac{1}{3} + \frac{158}{375R} - \frac{49}{625R^2}.$$

It is decreasing on $[2/5, 5/4]$, and $g(x) \geq \gamma(R)x^2/2$ whenever $|x| \leq R$. The largest possible $|\eta_i|$ in this sector is $19\pi/48 < 5/4$.

Split the maximum coordinate into the seventeen slabs

$$[\ell_j, R_j] = [2/5 + j/20,\ 2/5 + (j+1)/20], \qquad 0 \leq j \leq 16.$$

On a slab, use $K \leq H$ and the Gaussian precision $\gamma(R_j)V$. Its normalization costs at most $2(17/(16\gamma(R_j)))^{23}$. Equation (5.11) gives the explicit slab bound

$$64 \left(\frac{17}{16\gamma(R_j)} \right)^{23} e^{-t_j \ell_j \sqrt{\gamma(R_j)V} + 15t_j^2/32} \prod_{m=2,4,8,16} (1 + t_j^2/m). \tag{6.1}$$

At $V = 250$, take a rational lower bound for the square root by flooring it to denominator 1000, and set t_j equal to the integer part of the resulting $\ell_j \sqrt{250\gamma(R_j)}$, minus one, or one if this is smaller. Sum (6.1), bounding each negative exponential by the reciprocal of a positive exponential Taylor partial sum. The exact sum is less than $9/80$. Each term decreases for larger V. Thus the two principal central complements together contribute less than $9/40$.

6.2. Nonprincipal central sectors

The other central sectors have means $\mu = \pm 3\theta, \pm 5\theta$. Strong concavity gives

$$\sum_i h_s(\alpha_i) \leq 16 h_s(\mu) - \frac{v_0}{6} \sum_i (\alpha_i - \mu)^2.$$

The wide fourth-cumulant bound implies

$$h_s(\theta) - h_s(3\theta) \geq v_0(4\theta^2 - 10\theta^4) > v_0/8.$$

For example, integrate $\Re v(se^{ia}) \geq v_0(1 - 3a^2/2)$ first from zero to a, then from θ to 3θ. The larger mean 5θ only improves the loss. There are sixteen leaves, so the total mean deficit exceeds $2v_0$. The four sectors together are bounded by

$$8(51/16)^{23} e^{-2V} < 10^{-10}, \qquad V \geq 250. \tag{6.2}$$

6.3. Exactly one noncentral angle

Let its magnitude be $a \in [\pi/3, \pi]$. From (3.1),

$$h_s(a) \leq -v_0 \phi(a), \qquad \phi(a) = \frac{2 - \cos a - \cos^2 a}{3}.$$

For $|x| \leq 2\pi/45$, (3.7) and Taylor's formula give

$$h_s(x) \leq -v_0 x^2/2 + v_0 x^4/16 \leq -(49/100) v_0 x^2.$$

The mean of the fifteen inliers has magnitude at least $(\pi - a)/15$. Monotonicity of h_s, followed by the short-angle estimate at this smaller comparison argument, gives a mean cost at least $(49/1500)v_0(\pi - a)^2$. The remaining inlier fluctuations have Gaussian precision $V/3$.

The function $\phi(a) + (49/1500)(\pi - a)^2$ is bounded below by $5/12 + 49\pi^2/3375$. To check this, it is concave on $[\pi/3, 2\pi/3]$, because $\phi'' \le -1/6$ while the added second derivative is $49/750 < 1/6$; its first endpoint is the smaller one. On $[2\pi/3, \pi]$, $\phi \ge 2/3$, which is larger than that endpoint value. After subtracting (3.3), the deficit exceeds

$$\delta_1 = \frac{3246499}{13837500}, \tag{6.3}$$

using $\pi^2 > 493/50$.

Only the four ancestors of the outer leaf need their kernel factors bounded by two. The eleven remaining contrasts lie orthogonally in the fourteen-dimensional centered inlier space, and their product of $m_\nu/2$ is 32. Their Gaussian integral is $32(2\pi)^7(3/V)^{18}$. The original measure disintegrates as $da\, d\sigma/\sqrt{15}$. There are at most sixteen choices of leaf and six winding sectors. After normalizing by (4.1), the total is bounded by

$$128\, 3^{18}(17/16)^{23} V^{9/2} e^{-\delta_1 V} < 1/2000, \qquad V \ge 250. \tag{6.4}$$

The factor uses $\sqrt{2\pi/15} < 2/3$ and includes the conditioning factor two.

6.4. Exactly two noncentral angles

The two leaf-to-root paths have at most seven distinct internal ancestors. At least eight contrasts therefore involve only the fourteen central leaves. Keep eight such contrasts and bound the other seven kernel factors by two. Each retained node has size at most eight, so

$$K \le 2^{23} \prod_{j=1}^{8} x_j^2$$

in orthonormal coordinates of the thirteen-dimensional centered inlier plane. Its Gaussian integral contributes $(2\pi)^{13/2}(3/V)^{29/2}$. The coarea factor is $1/\sqrt{14}$, and there are at most $\binom{16}{2} \cdot 6 = 720$ leaf-pair/winding choices.

The deficit relative to the saddle is at least $2(5/12) - 40/123 = 125/246$. Including the conditioning factor two, the total bound is

$$720\, 2^{17} 3^{14} (17/16)^{23} V^8 e^{-125V/246} < 10^{-10}, \qquad V \ge 250. \tag{6.5}$$

Keeping the eight intact contrasts is essential: replacing the whole kernel by a constant would introduce a much larger power of V.

6.5. At least three noncentral angles

Here the deficit is at least $3(5/12) - 40/123 = 455/492$. Use $K \le 2^{15}$ and normalized total torus volume one. The resulting bound is

$$128(44/7)^8 (17/16)^{23} V^{23} e^{-455V/492} < 10^{-10}, \qquad V \ge 250. \tag{6.6}$$

No union multiplier is necessary: this is a pointwise bound on the entire region.

All numerical inequalities (6.1)–(6.6) are checked by rational arithmetic, positive exponential Taylor sums and elementary rational bounds on π. The functions $V^a e^{-\delta V}$ in (6.4)–(6.6) decrease for $V \ge 250$, as follows by differentiating their logarithms.

7. The uniform bulk theorem and phase points

Combining the two local errors, the two model-restoration errors, the actual complement, and (4.4), we obtain

$$\begin{aligned}\frac{|f_{n,4}(-s^{16}) - 2\Re A_n(s)|}{2|A_n(s)|} &< \frac{36821}{250^2} + \frac{53}{250} + \frac{11}{500} \\ &\quad + \frac{1}{2}\left(\frac{9}{40} + \frac{1}{2000} + \frac{3}{10^{10}}\right) \\ &= \frac{18717720003}{20000000000} < \frac{47}{50}\end{aligned} \tag{7.1}$$

whenever $0 < s \le 1$ and $V \ge 250$. Notice that the actual complement bounds were relative to one saddle; this accounts for the factor $1/2$ in the last term.

Define

$$\Psi_n(s) = 16n \arg p(\zeta) - \frac{45}{2} \arg v(\zeta).$$

Let $m_q = zq'/q$. Direct algebra gives, for $z = se^{i\theta}$,

$$\Im m(z) = \frac{s \sin\theta(1 + 4s\cos\theta + s^2)}{|p(z)|^2} > 0,$$

$$\Im m_q(z) = \frac{4s \sin\theta(1 + s\cos\theta + s^2)}{|q(z)|^2} \le 4\Im m(z).$$

The last inequality uses $|q|^2 - |p|^2 = 6s\cos\theta(1 + s^2) + 15s^2 \ge 0$. Since $zv'/v = 1 + m_q - 2m$,

$$\frac{d\Psi_n}{d\log s} \ge (16n - 45)\Im m(z) > 0 \qquad (n \ge 3). \tag{7.2}$$

Moreover,

$$\Psi_n(0+) = -45\pi/32, \qquad \Psi_n(1) = n\pi.$$

Thus every integer phase level $m\pi$, $0 \le m \le n$, has a unique point in $(0, 1]$. We use only levels $m \ge 250$, not formal phase crossings near the endpoint.

For $n \ge 1001$, let s_* solve $nv(s_*) = 250$. The real variance is strictly increasing on $(0, 1)$, since

$$sv'(s) = \frac{s(1 - s^2)(1 + 7s + s^2)}{p(s)^3} > 0.$$

The exact value $v(7/20) = 141260/346921$ shows $s_* < 7/20$. On this interval,

$$\frac{m(s)}{v(s)} - 1 = \frac{s(2s^2 + 2s - 1)}{1 + 4s + s^2} \le 0.$$

Also, for $0 \le \alpha \le \theta$,

$$\Im v(se^{i\alpha}) = \frac{s \sin\alpha(1 - s^2)(4s^2\cos^2\alpha + 8s^3\cos\alpha + 8s\cos\alpha + s^4 + 5s^2 + 1)}{|p(se^{i\alpha})|^4} \ge 0.$$

Therefore $\arg p(se^{i\alpha})$ is concave in α, and $\arg p(\zeta) \le \theta m(s)$, while $\arg v(\zeta) \ge 0$. Hence

$$\Psi_n(s_*) \le 16n\theta v(s_*) = 250\pi. \tag{7.3}$$

By monotonicity, all phase points $s_{n,m}$, $250 \le m \le n$, have $V \ge 250$. At such a point, $\Re A_n = (-1)^m |A_n|$, so (7.1) proves

$$(-1)^m f_{n,4}(-s_{n,m}^{16}) > 0, \qquad 250 \le m \le n, \quad n \ge 1001. \tag{7.4}$$

In particular, $s_{n,n} = 1$ and $f_{n,4}(-1) \ne 0$.

8. The analytic edge family and exact enclosure lemma

8.1. A family regular at $u = 0$

For complex u, z with $|uz| < 1$, define

$$g_u(z) = \exp L(u, z), \qquad L(u, z) = \begin{cases} \log p(uz)/u, & u \neq 0, \\ z, & u = 0. \end{cases}$$

The logarithm is its analytic branch at zero. The displayed singularity is removable. Set

$$b_{0,k}(u) = [z^k]g_u(z), \qquad b_{j+1,k} = b_{j,k}^2 - b_{j,k-1}b_{j,k+1}, \qquad F_u(x) = \sum_{k\geq 0} b_{4,k}(u)x^k. \tag{8.1}$$

These analytic functions are used only on domains where the bounds below ensure convergence. For a reciprocal integer $u = 1/n$,

$$b_{0,k}(1/n) = n^{-k}a_{n,k}, \qquad F_{1/n}(x) = f_{n,4}(x/n^{16}). \tag{8.2}$$

The second identity follows because one application of $\mathcal{L}$ squares a geometric coefficient scaling. It is exact, not an asymptotic limit.

8.2. Complex-parameter majorant

The series

$$\log p(w) = \sum_{\ell\geq 1} c_\ell w^\ell, \qquad c_\ell = \begin{cases} 1/\ell, & 3 \nmid \ell, \\ -2/\ell, & 3 \mid \ell, \end{cases}$$

follows from $p(w) = (1 - w^3)/(1 - w)$. Put

$$G(q) = -\log(1 - q) - \tfrac{1}{3}\log(1 - q^3), \qquad H(q) = \frac{qG'(q) - G(q)}{q^2}.$$

Let $c \geq 0$, $r, R > 0$, and $q = (c + r)R < 1$. On $|u - c| \leq r$, $|z| \leq R$,

$$|\partial_u L(u, z)| \leq R^2 H(q).$$

Integrating from c to u gives

$$|g_u(z)| \leq e^E, \qquad E = \frac{\log p(cR)}{c} + rR^2H(q), \tag{8.3}$$

where the quotient is interpreted as R when $c = 0$. At the real starting parameter $c > 0$, the bound follows from $|p(cz)| \leq p(cR)$.

Cauchy's coefficient estimate followed by four applications of $\mathcal{L}$ yields

$$|b_{4,k}(u)| \leq 2^{15}e^{16E}R^{-16k}. \tag{8.4}$$

Indeed the constants obey $C_{j+1} = 2C_j^2$, $C_0 = 1$, while the spatial exponent doubles at every layer. A missing negative-index neighbor only improves the estimate.

8.3. Moving-node Taylor remainder

Let a positive rational moving node on a real panel $[c-h, c+h]$ be

$$t(u) = \frac{t_c}{1+a(u-c)}, \qquad |a|r < 1, \qquad 0 < h < r.$$

Then $|t(u)| \le T = t_c/(1-|a|r)$ throughout the **complex** parameter disk. Choose $R > T$ with $(c+r)R < 1$. For the truncated evaluation

$$Q_K(u) = \sum_{k=0}^{K} b_{4,k}(u)(-t(u)^{16})^k,$$

(8.4) gives

$$|Q_K(u)| \le M = \frac{2^{15}e^{16E}}{1-(T/R)^{16}}.$$

If $Q_K(c+hw) = \sum_{j\ge 0} d_j w^j$, Cauchy's estimate proves

$$\left| Q_K(c+hw) - \sum_{j=0}^{d} d_j w^j \right| \le M\frac{(h/r)^{d+1}}{1-h/r}, \qquad |w| \le 1. \tag{8.5}$$

The finite coefficient truncation is bounded separately. For real $0 \le u \le U$,

$$\log p(uR_0)/u \le R_0 + UR_0^2/2 =: E_0,$$

because $\log p(q) \le q + q^2/2$. If $R_0 > T_0 = \max t(u)$ and $UR_0 < 1$, then

$$|F_u(-t(u)^{16}) - Q_K(u)| \le \frac{2^{15}e^{16E_0}(T_0/R_0)^{16(K+1)}}{1-(T_0/R_0)^{16}}. \tag{8.6}$$

The complex-disk radius in (8.5) need not equal the real coefficient-tail radius in (8.6). In the certificates, $r = 4h$, $R = T+1$, and $R_0 = 2T_0$ are used after checking the displayed domain inequalities exactly.

8.4. Exact finite Taylor arithmetic and the sign test

The retained Taylor coefficients are enclosed as

$$d_i \in 2^{e-P}[m_i - \rho, m_i + \rho], \qquad 0 \le i \le d,$$

with integer midpoints, a common nonnegative integer radius, and an arbitrary integer binary exponent. Thus rounding contributes at most $(d+1)\rho 2^{e-P}$ on $|w| \le 1$.

For multiplication, exact integer convolution of the midpoint vectors has raw coefficient error at most

$$\rho_b \sum |m_a| + \rho_a \sum |m_b| + (d+1)\rho_a\rho_b.$$

Its raw exponent is $e_a + e_b - P$. Rescaling a midpoint by a positive integer denominator D, with nearest integer n_i, enlarges the radius to

$$\left\lceil \frac{\rho + \max_i |Dn_i - m_i|}{D} \right\rceil.$$

These formulas prove the enclosure invariant independently of intermediate coefficient size. Normalization uses both midpoint magnitudes and the radius; a zero midpoint with nonzero radius is not discarded. For multiplication by a rational scalar N/D, $D > 0$, choose integer rounded midpoints n_i; a valid new common radius is

$$\left\lceil \frac{|N|\rho + \max_i |Dn_i - Nm_i|}{D} \right\rceil.$$

Addition first aligns binary exponents by these outward-rounded rescalings and then adds the radii. A formal coefficient shift preserves the common radius on retained entries. Thus all primitive operations used in the recurrence preserve the stated coefficientwise enclosure.

The initial jets are generated from the exact recurrence

$$b_{0,0} = b_{0,1} = 1, \qquad (k+1)b_{0,k+1} = (1-ku)b_{0,k} + u(2-(k-1)u)b_{0,k-1}. \tag{8.7}$$

Four radius-one stencils give the shared $b_{4,k}$ jets. At each moving node, Horner evaluation uses the Taylor jet

$$-t_c^{16}(1+ahw)^{-16}, \qquad [w^j] = -t_c^{16}\binom{15+j}{j}(-ah)^j.$$

Truncating after formal degree d cannot affect retained formal coefficients.

Convert the midpoint polynomial to Bernstein form after $w = 2x - 1$, $0 \le x \le 1$. Its minimum and maximum controls enclose all values. Add the rounding envelope and the two independent errors (8.5)–(8.6). A strictly positive lower bound or strictly negative upper bound then certifies the sign on the entire closed panel. This is an interval-wide proof, not a sampled evaluation.

All transcendental majorants can be replaced by rational upper bounds. For example, expand $H(q)$ through degree thirty and bound the remainder by $2q^{31}/(1-q)$; bound logarithms by the positive atanh series with its geometric tail; use a positive rational lower bound for $\log 2$ to replace e^X by $2^{\lceil X/\log 2_{\mathrm{lower}} \rceil}$. Consequently the final sign comparison is an exact rational inequality.

9. Parameter-wide bulk bypass and the final join

For real $r > 0$, the analytic function $p(z)^r$ on $|z| < 1$ is defined with the logarithm at zero. The diagonal identity, kernel symmetrization and all modulus estimates remain valid: integrality of r was not used in those steps. Thus (7.1) extends to real r, $0 < s < 1$, $rv(s) \ge 250$. No polynomial degree or reciprocal-root assertion is made for noninteger r.

At a moving node, set $r = 1/u$, $s = ut(u)$. If

$$rv(s) \ge 250, \qquad |\Psi_r(s) - j\pi| \le 1/3,$$

then (7.1) gives the desired sign because $\cos(1/3) \ge 17/18 > 47/50$. At $u = 0$, the analytic family tends uniformly on compact sets to its Poisson-scale limit $g_0(z) = e^z$. The amplitude has a finite positive limiting magnitude; the strict uniform margin therefore extends the sign to that endpoint.

These bypass inequalities and the final join are themselves low-degree rational polynomial certificates. Write $t = A/B$, $B > 0$, and let $c = \cos\theta, d = \sin\theta, c_2 = \cos 2\theta, d_2 = \sin 2\theta$. Then

$$\arg p(ute^{i\theta}) = \arctan \frac{uN_p}{D_p}, \quad D_p = B^2 + cuAB + c_2u^2A^2, \quad N_p = dAB + d_2uA^2,$$

with $D_q = B^2 + 4cuAB + c_2u^2A^2$, $N_q = 4dAB + d_2uA^2$ for q. Hence

$$\Psi_{1/u}(ut) = (16/u + 45)\arctan(uN_p/D_p) - (45/2)\arctan(uN_q/D_q) - 45\pi/32. \tag{9.1}$$

All real parts are positive, so the branches are unambiguous. Use certified rational sine, cosine and π enclosures, together with

$$x - x^3/3 \le \arctan x \le x - x^3/3 + x^5/5 \qquad (x \ge 0).$$

The upper polynomial is increasing; the lower is increasing on $[0, 1]$, which is verified for the relevant ratio bounds. After division by u, the apparent singularity in (9.1) is removable. Clearing positive denominators gives degree-sixteen polynomial phase margins. The variance condition is the degree-four inequality obtained from

$$\frac{v(ut)}{u} = \frac{AB(B^2 + 4uAB + u^2A^2)}{(B^2 + uAB + u^2A^2)^2}. \tag{9.2}$$

Positive Bernstein controls certify both types of margin on each whole panel.

For the supplied final harmonic node, the exact inequalities establish $\Psi_{1/u}(ut_{249}(u)) < 250\pi$. For the root argument we use this only at reciprocal integers $u = 1/n$: phase monotonicity then gives

$$t_{249}(1/n) < ns_{n,250}. \tag{9.3}$$

10. Certified edge inputs and assembly

The rational parameter grid is

$$u_\ell = \ell/32032, \qquad 0 \le \ell \le 32.$$

For each $j = 1, \dots, 249$, the endpoint table supplies positive rational values $T_{j,\ell}$. On a panel interpolate reciprocals:

$$t_j(u) = \left(\frac{1-\lambda}{T_{j,\ell}} + \frac{\lambda}{T_{j,\ell+1}}\right)^{-1}, \quad \lambda = \frac{u - u_\ell}{u_{\ell+1} - u_\ell}.$$

The exact adjacent endpoint comparisons imply strict node ordering throughout every panel: reciprocal interpolation preserves the reversed inequalities, and the final reciprocal restores their order.

The completed finite edge certificate consists of panels $0, \dots, 7$, exactly covering

$$[0, 8/32032] = [0, 1/4004].$$

On each of these panels, every node $j = 1, \dots, 249$ has the strict sign

$$(-1)^j F_u(-t_j(u)^{16}) > 0. \tag{10.1}$$

The certificate inputs are as follows; the entries distinguish Taylor enclosures from the exact bulk-bypass criterion, not separate mathematical conclusions.

Panel	Taylor-enclosed signs	Bulk-bypass signs	Total
0	246	3	249

Panel	Taylor-enclosed signs	Bulk-bypass signs	Total
1	245	4	249
2	243	6	249
3	241	8	249
4	240	9	249
5	239	10	249
6	238	11	249
7	237	12	249

Each Taylor entry includes an exact final coefficient enclosure and strict rational comparison against both analytic errors. The hypotheses of (8.3)–(8.6) are checked entry by entry. The bypass entries use the exact inequalities in §9. The final join (9.3) and ordering use the same frozen rational node table. None of these statements follows merely from the aggregate counts in the table.

Now fix an integer $n \geq 4004$. Its reciprocal lies in the certified closed parameter union. Since $F_{1/n}(0) = 1$, the first 249 alternating edge signs give 249 disjoint root intervals. The last edge sign is negative, while phase point 250 has positive sign, so (9.3) supplies one more interval. Bulk phases 250 through n supply $n - 250$ further intervals. In the original polynomial variable these are

$$249 + 1 + (n - 250) = n$$

disjoint intervals inside $(-1, 0)$. Thus $f_{n,4}$ has at least n distinct roots there.

The operator $\mathcal{L}$ preserves palindromicity, degree $2n$, and both endpoint coefficients one. Reciprocal roots give another n distinct roots below -1. The value at -1 is nonzero by (7.4). These $2n$ roots exhaust the degree, so every root is strictly negative and real.

The companion finite/algebraic chapter proves nonnegativity of iterates zero through three for $n \geq 56$, by finite PF4 seed certificates and convolution closure. This includes the entire infinite range used here. Brändén's theorem then gives all iterates from four onward. For $10 \leq n \leq 4003$, exact directed ratio certificates enter the invariant cone $\max q_k \leq 1/3$, which implies all subsequent iterates are positive. The finite and infinite ranges meet without a gap and cover every integer $n \geq 10$.

11. Reproducibility boundary

The mathematical proof requires more than a floating plot or a list of successful samples. Its finite data are:

- the rational Bernstein certificates for the scalar inequalities in §3;
- the exact weighted-moment sum in §5;
- the rational complement bounds in §6;
- the frozen moving-node endpoint table, interval-wide sign enclosures, and final-join controls;
- the finite cone and PF4 certificates described in the companion chapter.

The computation uses reviewed exact integer/rational arithmetic and outward-rounded coefficient enclosures. An executable certificate checker must verify domains, strict sign inequalities, index coverage and the connection between every saved enclosure and its parameters. Cryptographic hashes help identify the intended records; hashing alone does not establish any mathematical inequality. This draft presents a computer-assisted proof, not a claim of formal proof-assistant verification or external peer review.

Primary references

1. L. A. Medina and A. Straub, *On multiple and infinite log-concavity*, 2014 preprint; Annals of Combinatorics **20** (2016), 125–138. Author-hosted paper, especially Example 5.3 and Table 1.
2. P. Brändén, *Iterated sequences and the geometry of zeros*, Journal für die reine und angewandte Mathematik **658** (2011), 115–131. arXiv version, especially §3.

Reproducing the computational premises

The accompanying `certificate_package` has three independent directories: `finite`, `analytic`, and `edge`. It requires Python with assertions enabled. Exact analytic polynomial checks use SymPy 1.14.0, and accelerated edge regeneration uses python-flint 0.9.0. The finite checks and saved-edge replay require only the standard library. Dependency versions and backend provenance are pinned in the package.

From the package directory, the following commands reconstruct the stated premises:

```
python finite/verify_pf4.py
python finite/verify_finite.py --full --output finite_result.json
python analytic/run_analytic.py --full
python edge/regenerate.py --full
python edge/replay_saved_inequalities.py
```

The finite command verifies all 3994 exponents in the finite range, not a selection. The analytic command reconstructs the rational polynomial and weighted-moment arithmetic underlying the displayed inequalities. The edge-generation command reconstructs all interval enclosures from the coefficient recurrence and four applications of the operator. Finally, the saved-inequality command evaluates every enclosure's strict sign margin and its two analytic remainder budgets.

For a shorter inspection, `python edge/replay_saved_inequalities.py` alone checks all 1929 saved enclosure inequalities, 63 bypass boxes, node ordering and the final join. This shorter command is deliberately described as *conditional on the saved enclosures*: it does not reproduce their interval-propagation provenance. The separate representative command

```
python edge/regenerate.py --panel 0 --node 200
```

reconstructs one enclosure and compares every deterministic midpoint, radius and exponent. A successful representative test is not represented as a full-chain rerun. The original complete generation and independent saved-inequality checks were performed; the portable full-generation command is supplied for reproduction.

Eight deterministic compressed data files contain the original final jets, one file per parameter panel. Hash manifests authenticate the files and identify the exact reproducibility version. Hash agreement is not a substitute for the mathematical implications proved in the manuscript or for arithmetic reconstruction. Intermediate shared coefficient checkpoints are omitted from the compact distribution; full regeneration rebuilds them in memory.

The computational evidence and proof composition have received independent internal checks. External peer review and proof-assistant formalization remain distinct activities; neither is claimed by the present draft.

References

1. L. A. Medina and A. Straub, *On multiple and infinite log-concavity*, arXiv:1405.1765v1 (2014); *Annals of Combinatorics* **20** (2016), 125–138. The specific target is Example 5.3, printed page 11 of the preprint. https://arxiv.org/abs/1405.1765v1
2. P. Brändén, *Iterated sequences and the geometry of zeros*, arXiv:0909.1927v3 (2010); *Journal für die reine und angewandte Mathematik* **658** (2011), 115–131. Section 3 gives the real-nonpositive-zero preservation used after the fourth iterate. https://arxiv.org/abs/0909.1927v3
3. P. R. W. McNamara and B. E. Sagan, *Infinite log-concavity: developments and conjectures*, *Advances in Applied Mathematics* **44** (2010), 1–15; arXiv:0808.1065. The symmetric-function iteration identities and Toeplitz positivity provide context for the earlier-layer argument. https://arxiv.org/abs/0808.1065
4. I. G. Macdonald, *Symmetric Functions and Hall Polynomials*, second edition, Oxford University Press, 1995, Chapter I, Sections 3, 5 and 9. Used for dual Jacobi–Trudi and Littlewood–Richardson identities.

The exact polynomial-multiplication implementation uses the released python-flint 0.9.0 interface to FLINT. This is an arithmetic dependency, not a mathematical theorem proved here. Its release provenance and exact conversion contract are documented with the verification software. https://pypi.org/project/python-flint/0.9.0/